

THE SUPER-RICCATI SHADOW TOWER AND THE VERDIER COMPLEMENTARITY PROBLEM

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ABSTRACT. The super-Riccati tower gives a theorem for shadow coefficients attached to a homogeneous primary line after the line-restricted Maurer–Cartan equation is put in the Riccati normalisation fixed below. The seed data S_3, S_4 are external OPE data; the recurrence starts at $r = 5$. The parity of the line first changes the diagonal $(4, 4)$ -term in S_6 . The Verdier complementarity statement for $Y(\mathfrak{sl}(m|n))$ is separate from the recurrence. The rank-two calculation in the stated super-trace normalisation gives $3 = \dim \mathfrak{sl}(2)$. Its conjectural family extension has constant $\dim \mathfrak{sl}(M) = M^2 - 1$, where $M = \max(m, n)$. The uniform scalar convention with numerator M instead has constant M . The Berezinian normalisation has the distinct conjectural constant $|m - n|$ only after the Berezinian–Verdier datum is fixed. Shadow coefficients and Igusa denominator scalars are coefficient data; they do not construct compact Hall objects, quasi-NCCR realisations, Hall–Drinfeld doubles, orientation systems, PBW bases, or compact CoHA.

I. RICCATI RECURRENCE

Definition 1 (Riccati-normalised primary line). Fix a homogeneous primary line ℓ with parity $|\ell|$ and nonzero coefficient κ_ℓ . Let S_r be the shadow coefficient of degree r , and choose line descendants θ_r with

$$|\theta_r| = (r - 1)|\ell| \pmod{2}.$$

Set

$$f(j, k) = \begin{cases} 1, & j < k, \\ \frac{1}{2}, & j = k. \end{cases}$$

The line is Riccati-normalised if the degree- r component ($r \geq 5$) of the line-restricted Maurer–Cartan equation is

$$2r\kappa_\ell S_r + \sum_{\substack{j+k=r+2 \\ 3 \leq j \leq k < r}} (-1)^{(j-1)(k-1)|\ell|} f(j, k) j k S_j S_k = 0.$$

Here $f(j, k)$ is the unordered Wick-pair factor obtained after the ordered quadratic term $\frac{1}{2}[\Theta_\ell, \Theta_\ell]$ is quotiented to $j \leq k$: off-diagonal pairs occur once, while diagonal pairs carry the residual $\frac{1}{2}$ symmetry factor in this normalisation.

Definition 2 (Admissible seed). An admissible seed for a Riccati-normalised line ℓ is a pair (S_3, S_4) obtained from the line-restricted OPE data, satisfying locality, parity, and non-degeneracy of the Zamolodchikov pairing $\kappa_\ell \neq 0$. These two scalars are not produced by the recurrence. If ℓ is a pure single-mode odd line in a parity-preserving theory, the scalar cubic coefficient satisfies $S_3 = 0$. The quartic seed S_4 vanishes only when the chosen line has no genuine weight-four even composite channel; otherwise it is independent input.

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Theorem 3 (Super-Riccati recurrence). *For a Riccati-normalised primary line with admissible seed pair (S_3, S_4) , the coefficients for $r \geq 5$ satisfy*

$$S_r = -\frac{1}{2r\kappa_\ell} \sum_{\substack{j+k=r+2 \\ 3 \leq j \leq k < r}} (-1)^{(j-1)(k-1)|\ell|} f(j, k) j k S_j S_k.$$

Proof. The Riccati normalisation says precisely that the degree- r component of the line-restricted Maurer–Cartan equation is

$$2r\kappa_\ell S_r + \sum_{\substack{j+k=r+2 \\ 3 \leq j \leq k < r}} (-1)^{(j-1)(k-1)|\ell|} f(j, k) j k S_j S_k = 0$$

with the Koszul sign computed from $|\theta_j||\theta_k| = (j-1)(k-1)|\ell|$. The coefficient $2r\kappa_\ell$ is nonzero because $\kappa_\ell \neq 0$ and the base field has characteristic zero. Solving the displayed equation for S_r gives the recurrence. $\square$

Remark 4. On a pure odd line the cubic seed satisfies $S_3 = 0$. The quartic seed S_4 is non-recursive OPE data and may vanish depending on the chosen line.

Proposition 5 (First recursive coefficients). *For every Riccati-normalised line,*

$$S_5 = -\frac{6}{5\kappa_\ell} S_3 S_4$$

and

$$S_6 = -\frac{1}{12\kappa_\ell} \left(15 S_3 S_5 + 8(-1)^{|\ell|} S_4^2 \right).$$

Equivalently, in seed-only form,

$$S_6 = \frac{S_4}{6\kappa_\ell^2} \left(9 S_3^2 - 4(-1)^{|\ell|} \kappa_\ell S_4 \right).$$

Thus the even-line recurrence is the bosonic Riccati recurrence, while the first parity-dependent recursive sign occurs in the diagonal $(4, 4)$ contribution to S_6 . The values through S_6 are normalisation fingerprints for the chosen line; they do not determine a full shadow class, a compact source, or a Verdier datum without the seed and line normalisation.

Proof. For $r = 5$, the only pair is $(j, k) = (3, 4)$, and $(-1)^{(2)(3)|\ell|} = 1$, so

$$S_5 = -\frac{1}{10\kappa_\ell} 12 S_3 S_4 = -\frac{6}{5\kappa_\ell} S_3 S_4.$$

For $r = 6$, the pairs are $(3, 5)$ and $(4, 4)$. The first has sign $(-1)^{(2)(4)|\ell|} = 1$ and coefficient 15. The second has coefficient $f(4, 4) \cdot 16 = 8$ and sign $(-1)^{(3)(3)|\ell|} = (-1)^{|\ell|}$. Substitution into the recurrence gives the displayed formula.

Equivalently, the two recursive levels are the following term-by-term expansions:

r	(j, k)	$(-1)^{(j-1)(k-1) \ell } f(j, k) j k$	contribution
5	$(3, 4)$	12	$12 S_3 S_4$
6	$(3, 5)$	15	$15 S_3 S_5$
6	$(4, 4)$	$8(-1)^{ \ell }$	$8(-1)^{ \ell } S_4^2$

Substituting the displayed value of S_5 gives the seed-only formula. $\square$

2. SHADOW DATA AND HALL OBSTRUCTIONS

Definition 6. The shadow datum of the primary line is the scalar sequence

$$\text{Sh}_\ell(A) = (S_2, S_3, S_4, S_5, \dots), \quad S_2 := \kappa_\ell, \in k^{\mathbb{N}_{\geq 2}}.$$

The obstruction tuple for a compact Hall source is

$$\mathbf{o}_{\text{Hall}} = (\mathbf{o}_{\text{or}}, \mathbf{o}_{\text{corr}}, \mathbf{o}_{\text{coprod}}, \mathbf{o}_{\text{pair}}, \mathbf{o}_{\text{rad}}, \mathbf{o}_{\text{PBW}}, \mathbf{o}_{\text{comp}}),$$

where the entries ask for an orientation/Pfaffian line, compact Hall correspondences, a coproduct, a non-degenerate Hopf pairing, radical descent, PBW/no-extra-relations compatibility, and compact support.

Proposition 7. *The recurrence for $\text{Sh}_\ell(A)$ constructs no compact Hall object, orientation datum, PBW theorem, or compact CoHA:*

$$\text{Sh}_\ell(A) \not\Rightarrow \mathbf{o}_{\text{Hall}} = \mathbf{o}.$$

Proof. The map $A \mapsto \text{Sh}_\ell(A)$ lands in a sequence of scalars computed from one primary line. Compact Hall data live on a moduli problem with extension correspondences and vanishing-cycle sheaves. PBW requires a filtered algebra and an associated-graded comparison. Orientation requires a square root of the determinant/Pfaffian line. None of these structures is present in the codomain $k^{\mathbb{N}_{\geq 2}}$, and no map from the displayed scalar sequence to the obstruction tuple is constructed by the Riccati recurrence. A scalar match can be a necessary compatibility check; it is not a construction of the Hall source and not a vanishing proof for $\mathbf{o}_{\text{Hall}}$. $\square$

Proposition 8 (Igusa scalar normalisation). *When the $K_3 \times E$ denominator is compared with a shadow tower, the physical unnormalised genus-two scalar, the OP square, and the OP/DT reciprocal scalar are*

$$\begin{aligned} \Phi_{10}^{\text{un}} &= \Delta_5^2, & D_5 &= 64^{-1} \Delta_5, & \Phi_{10}^{\text{OP}} &= D_5^2 = 4096^{-1} \Delta_5^2, \\ Z_{\text{OP/DT}} &= -(\Phi_{10}^{\text{OP}})^{-1} = -D_5^{-2} = -4096 \Delta_5^{-2}. \end{aligned}$$

The equality $D_5 = 64^{-1} \Delta_5$ is the monic-versus-theta normalisation of the Gritsenko–Nikulin product; its square gives $\Delta_5^2/4096$. The minus sign belongs to the reciprocal OP/DT generating function, not to the Borchers product. These scalar identities do not construct a compact critical CoHA, a quasi-NCCR realisation, a Hall–Drinfeld double, or a vanishing proof for $\mathbf{o}_{\text{Hall}}$.

Proof. Let D_5 be the monic Borchers product attached to the $\phi_{0,1}$ -lift in the standard Gritsenko–Nikulin sign convention. The theta-normalised form has leading coefficient $[q^{1/2} r^{1/2} s^{1/2}] \Delta_5 = 64$, while D_5 is monic. Hence $D_5 = 64^{-1} \Delta_5$, and

$$D_5^2 = (64^{-1} \Delta_5)^2 = 4096^{-1} \Delta_5^2.$$

The physical unnormalised convention keeps the unscaled theta-constant normalisation and therefore writes $\Phi_{10}^{\text{un}} = \Delta_5^2$. The Oberdieck–Pandharipande scalar convention uses the monic product, hence $\Phi_{10}^{\text{OP}} = D_5^2$, and the reduced OP/DT generating function is the reciprocal with its separate sign convention: $-(\Phi_{10}^{\text{OP}})^{-1} = -4096 \Delta_5^{-2}$. This computation changes only the scalar normalisation. It supplies no orientation line, no compact extension correspondence, no monoidal vanishing-cycle realisation, no Hopf pairing, no radical quotient, no PBW filtration comparison, no quasi-NCCR realisation, no Hall–Drinfeld double, and no compact-CoHA construction. $\square$

3. COMPLEMENTARITY

Definition 9 (Principal-line scalar conventions). Let $m, n \geq 1$, $m \neq n$, $M = \max(m, n)$, and $h = |m - n|$. The dimension-normalised principal even line is the scalar convention

$$c_{\dim}(k) = \frac{(M^2 - 1)k}{k + h}, \quad \kappa_{\text{ch}}^{\dim}(k) = \frac{1}{2}c_{\dim}(k).$$

The uniform principal-line convention is

$$c_{\text{unif}}(k) = \frac{Mk}{k + h}, \quad \kappa_{\text{ch}}^{\text{unif}}(k) = \frac{1}{2}c_{\text{unif}}(k).$$

Both use the Sugawara-shift inversion $k^\vee = -k - 2h$. They are equal only in the accidental ranks solving $M = M^2 - 1$, and no such rank has $M \geq 2$.

Proposition 10 (Complementarity constants). *For the two scalar conventions above,*

$$\kappa_{\text{ch}}^{\dim}(k) + \kappa_{\text{ch}}^{\dim}(k^\vee) = M^2 - 1, \quad \kappa_{\text{ch}}^{\text{unif}}(k) + \kappa_{\text{ch}}^{\text{unif}}(k^\vee) = M.$$

Proof. For any constant C and $h \neq 0$,

$$\frac{1}{2} \frac{Ck}{k + h} + \frac{1}{2} \frac{C(-k - 2h)}{-k - h} = \frac{Ck + C(k + 2h)}{2(k + h)} = C.$$

Taking $C = M^2 - 1$ gives the dimension-normalised constant. Taking $C = M$ gives the uniform constant. $\square$

Remark 11. The preceding proposition is an algebraic identity for two scalar conventions. It does not identify either scalar convention with the principal even line of $Y(\mathfrak{sl}(m|n))$. The rank-two computation below is proved in the stated normalisation; the family statement with constant $M^2 - 1$ is conjectural until the dimension-normalised sub-Sugawara central-charge formula and Verdier level-duality involution are constructed.

Conjecture 12 (Dimension-normalised super-trace complementarity). *The principal even sub-Sugawara line of $Y(\mathfrak{sl}(m|n))$ carries a Verdier-compatible dimension normalisation. In that normalisation,*

$$\kappa_{\text{ch}}^{\text{str}}(Y(\mathfrak{sl}(m|n))) + \kappa_{\text{ch}}^{\text{str}}(Y(\mathfrak{sl}(n|m))^!) = M^2 - 1.$$

For $M \geq 3$ this requires a rank-uniform sub-Sugawara central-charge formula with numerator $M^2 - 1$ and a compatible Verdier level-duality involution. The uniform convention with numerator M proves the different constant M .

Proposition 13 (Rank-two super-trace check). *Assume the $Y(\mathfrak{sl}(2|1))$ principal even sub-Sugawara line has*

$$c_T(k) = \frac{3k}{k + 1}, \quad \kappa_{\text{ch}}^{\text{str}}(k) = \frac{c_T(k)}{2},$$

and Verdier level inversion $k^\vee = -k - 2$. Then

$$\kappa_{\text{ch}}^{\text{str}}(k) + \kappa_{\text{ch}}^{\text{str}}(k^\vee) = 3 = \dim \mathfrak{sl}(2).$$

Proof. The inverted central charge is

$$c_T(-k - 2) = \frac{3(-k - 2)}{-k - 1} = \frac{3(k + 2)}{k + 1}.$$

Hence

$$\frac{1}{2}c_T(k) + \frac{1}{2}c_T(-k - 2) = \frac{3k + 3(k + 2)}{2(k + 1)} = 3.$$

The parity-swapped $Y(\mathfrak{sl}(1|2))$ line has the same scalar normalisation. $\square$

Definition 14 (Berezinian–Verdier datum). A Berezinian–Verdier datum for $Y(\mathfrak{sl}(m|n))$ is the tuple

$$\mathcal{V}_{m|n} = (\mathfrak{X}_{m|n}, \text{Ber}_{\mathfrak{X}_{m|n}}, \delta_{m|n}, \mathbb{D}_{\text{super}}, \langle -, - \rangle_{\text{Ber}}, \iota_{m|n}, \lambda_{\text{Ber}}).$$

Here $\mathfrak{X}_{m|n}$ is a chosen super moduli object carrying the Yangian-side $\mathcal{D}$ -modules, $\text{Ber}_{\mathfrak{X}_{m|n}}$ is its Berezinian dualising line, $\delta_{m|n} \in \mathbb{Z}$ is the cohomological shift included in the datum, and

$$\mathbb{D}_{\text{super}}(\mathcal{M}) = \text{RHom}(\mathcal{M}, \text{Ber}_{\mathfrak{X}_{m|n}})[\delta_{m|n}]$$

is the Verdier functor. The pairing $\langle -, - \rangle_{\text{Ber}}$ is non-degenerate on the principal line, $\iota_{m|n}$ is the level-duality involution, and λ_{Ber} is the scalar extraction rule defining $\kappa_{\text{ch}}^{\text{Ber}} = \lambda_{\text{Ber}}(\langle -, - \rangle_{\text{Ber}})$. Without this whole datum the symbol $\kappa_{\text{ch}}^{\text{Ber}}$ is undefined.

Conjecture 15 (Berezinian normalisation). *Given a Berezinian–Verdier datum $\mathcal{V}_{m|n}$, the Berezinian complementarity conjecture is*

$$\kappa_{\text{ch}}^{\text{Ber}}(Y(\mathfrak{sl}(m|n))) + \kappa_{\text{ch}}^{\text{Ber}}(Y(\mathfrak{sl}(n|m)))^! = |m - n|$$

after the duality involution $\iota_{m|n}$, the pairing $\langle -, - \rangle_{\text{Ber}}$, the shift $\delta_{m|n}$, and the scalar map λ_{Ber} are fixed. This is a conjecture about the Berezinian–Verdier datum. It is not a consequence of the super-Riccati recurrence and is independent of the super-trace constant $M^2 - 1$. A numerical value $|m - n|$ is therefore not defined until the datum $\mathcal{V}_{m|n}$ and the scalar extraction rule λ_{Ber} are part of the statement.

Remark 16. The Verdier datum is the super- $\mathcal{D}$ -module category or formal moduli object, the duality functor $\mathbb{D}_{\text{super}}$, the Berezinian dualising line, the Berezinian pairing, the level-duality convention, and the scalar extraction rule. Any rank-wise compatibility with the super-Riccati coefficients is an additional comparison condition, not part of the recurrence.